

1. Simply observing a correlation between two variables X and Y is not enough to conclude that X has a causal effect on Y . When X and Y are correlated, there are several possible cases. Explain these cases with specific examples.
2. Explain the linear regression model in the example of class size and student performance. What are the parameters of the model? How to interpret these parameters? What is the meaning of the error term?
3. A researcher runs a randomized experiment to measure the impact a short nap has on memory. 200 participants in the sample are allowed to take a short nap of either 60 minutes or 75 minutes. After waking up, each of the participants takes a short test on short-term recall. Each participant is randomly assigned one of the examination times. Let Y_i denote the number of points scored on the test by the i -th participant, and let X_i denote the indicator variable for the amount of time for which the participant slept prior to taking the test ($X_i = 0$ if the i -th participant slept for 60 minutes and $X_i = 1$ for 75 minutes). Consider the linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$.
 - (a) Explain what the term u_i represents. Why will different students have different values of u_i ?
 - (b) Explain the meaning of $E[u_i]$ and $E[u_i|X_i]$ in this experiment.
 - (c) Do you expect that in this experiment, $E[u_i] = E[u_i|X_i]$? Explain.
4. Let X and Z be two independently distributed standard normal random variables, and let $Y = X^2 + Z$.
 - (a) Show that $E[Y|X] = X^2$.
 - (b) Show that $E[Y] = 1$.
 - (c) Show that $E[XY] = 0$.
5. Explain the Type I and Type II errors and the trade-off between the two errors by using a specific example.